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Partnership Prospectus · Humanoid Robotics for the Home · NSF SBIR Phase I lane

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PRIVATE CIRCULATION - grant and partnership lane only. Prepared for NSF program officers, prospective university partners, and specialist collaborators.

THE VALUE PROPOSITION

A US small business has designed a complete humanoid robotics platform for the one market no manufacturer serves - the home - at a bill of materials priced at 9.6-31.8% of one commercial-grade humanoid, with a full-body engineering specification, a costed build, a mapped funding pipeline, and a research record that is version-controlled and verifiable commit by commit. The company seeks a Phase I partnership to take the platform from funded conceptualization to prototype: NSF SBIR Phase I (up to $305,000), with Phase II ($1.25M) and university collaboration lanes already scoped.

THE GAP, WITH RECEIPTS

> **140+ companies build humanoids for factories. None builds for homes. No humanoid on the market in 2026 is ready for home use.**

- Field census (web-verified October 2026): 140+ humanoid manufacturers; roughly 7,000-8,000 commercial humanoids operating; industry concentration in factory settings; flagship consumer programs unpurchasable or factory-targeted.
- The company's costed bill of materials prices at 9.6-31.8% of one commercial-grade humanoid ($200K class), with arithmetic shown in-document - a design objective the field does not hold.
- The platform carries a full-body engineering specification (v3.0 master manual): 26 numbered design laws, 17 system sections, thermal/power/sensory/self-repair subsystems, and a 291-source bibliography compiled by script from the build corpus.

THE FOUNDER

Catherine Robinson-Rutella - physicist, robotics systems architect, and business developer. Ten years of full-cycle business development before the lab; independent research program (Stardust Lab) since March 2026.

- **The record:** 697 pressed technical documents, 213 print editions, 58 tool systems, 25 original research papers, and 1,212 version-control commits in 33 days - every commit hash verifiable on demand.
- **The speed, timestamped:** a complete investor package produced in 50.0 minutes of elapsed build time and a robotics build chain in 28.5 minutes, both proven from path-verified git timestamps. The full body specification (18 gated design documents) was produced in a single overnight program.
- **The discipline:** every derived number carries its arithmetic; every claim traces to a receipt; the record survives its own audit.
- Independent Research Program, Stardust Lab (2026); Fordham University, psychology coursework (2024-2025); research corpus published under the pen name Cici Pink.

WHY THIS FITS NSF PHASE I

- **The funded stage is exactly this stage.** NSF SBIR Phase I funds proof of concept - technical merit, feasibility, and commercial potential. The company's conceptualization package (specification v3.0, costed BOM, feasibility arithmetic, grant roadmap) is the Phase I work product already in hand, ready to be defended and extended.

- **The amounts and clock:** Phase I up to $305,000 (6-18 months); Phase II up to $1.25M; next full-proposal deadline November 4, 2026, gated by a mandatory Project Pitch (solicitation NSF 26-510). Fast-Track awards combine both phases.
- **Eligibility holds:** US small business, under 500 employees, majority US-citizen owned; no VC majority ownership.
- **Ranked topic fits already mapped:** NSF SBIR Robotics topic (best overall), NIDILRR SBIR (disability products - the platform's human-experience thesis matches the topic sentence for sentence), AFWERX humanoid topics ($2M class), NRI-3.0 ($250K-$1.5M, institutions only - the university-partner route this document proposes).

THE ASK - WHAT PARTNERSHIP LOOKS LIKE

- **NSF lane:** collaboration on the Project Pitch and Phase I proposal for the November 4, 2026 deadline; the company carries the design record, the partner carries complementary validation and specialized capability.
- **University lane (NRI-3.0):** a research partnership placing the platform's specification and methods inside an institutional proposal; the company's human-experience methodology (organ-ledger design, 1:1 experiential mapping) is a publishable collaboration surface.
- **Specialist lane:** hardware, controls, and biomechanics collaborators for the prototype stage, scoped in the company's collaborator paper.
- **Terms:** deliverables are licensed, never assigned; methods and architectures stay with the practice. Engagement structure and rate card available on request.

VERIFICATION

Every claim above traces to a commit hash, a script-printed disk count, a dated external source, or a published artifact in a version-controlled record, and is provable from company output on request. Field figures (manufacturer counts, sector funding, award amounts) are web-verified October 2026 with sources cited in the underlying documents. A verification appendix mapping each claim to its receipt is available on request.

Companion documents: full-body specification manual (v3.0); investor evaluation with git-timestamp proofs; costed bill of materials and feasibility arithmetic; grants roadmap; collaborator paper. Full-lab valuation available privately.